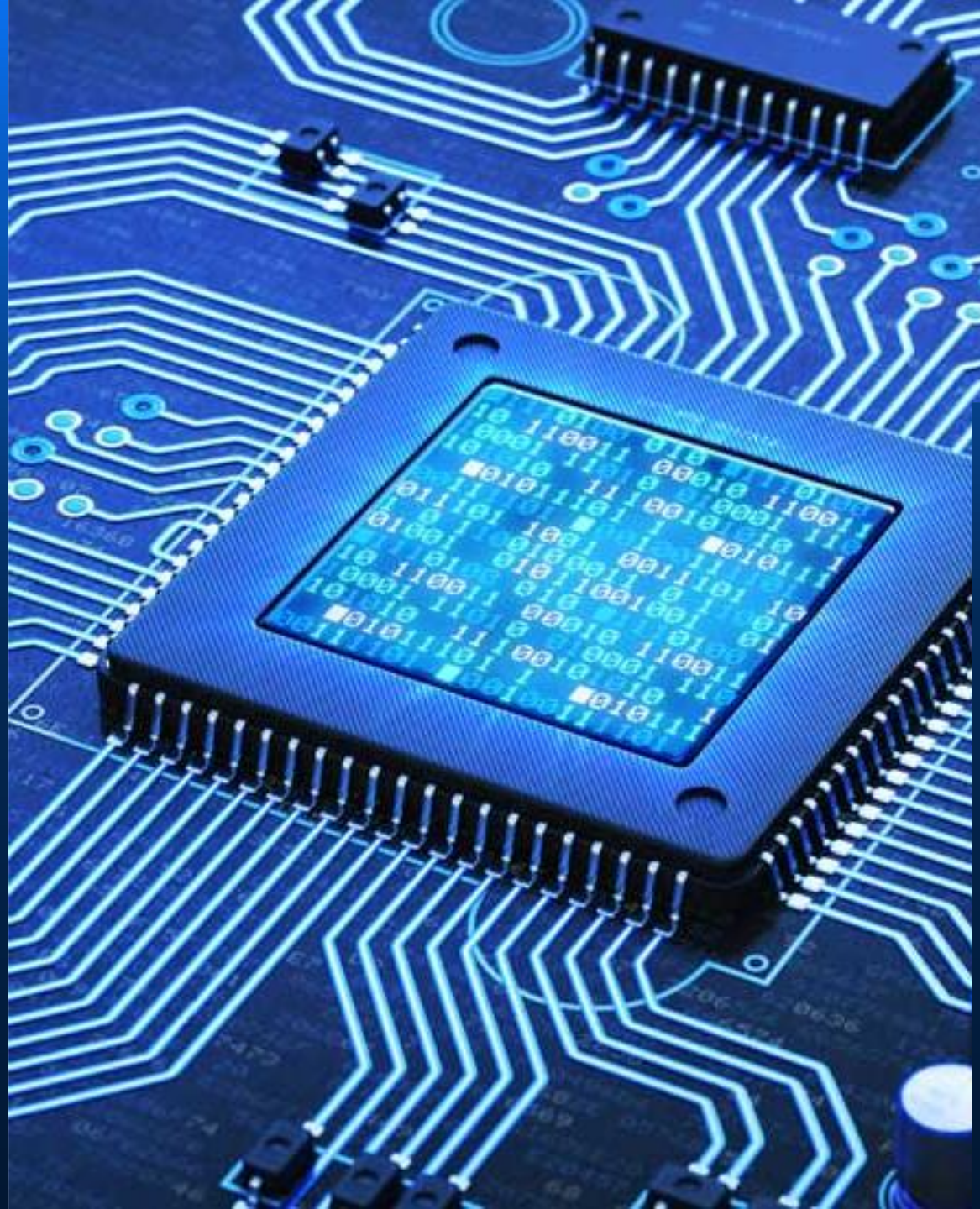


Computer Architecture

Session 10

By: Eng. Abdulkareem Mohammed

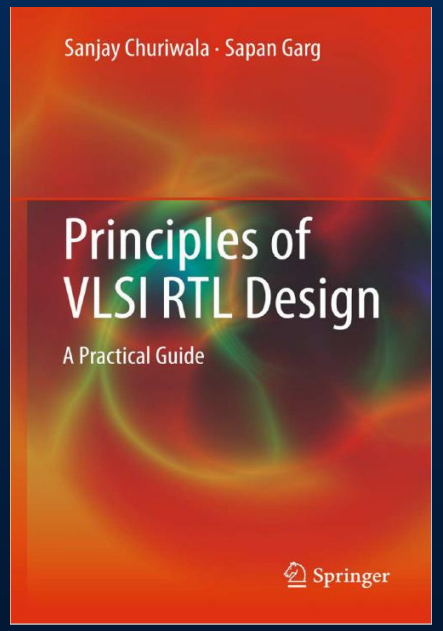
Senior HW Digital Design Engineer

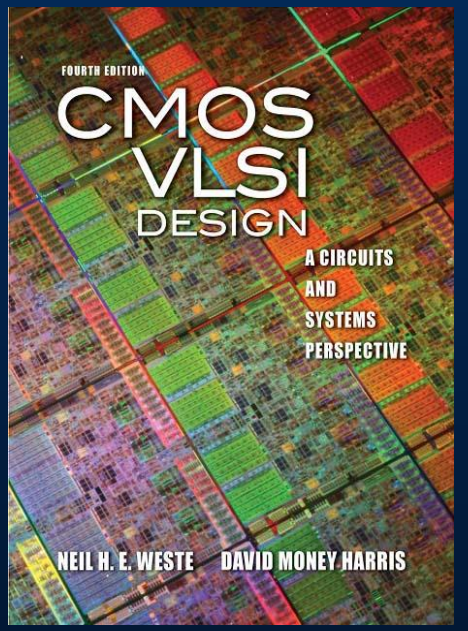
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This presentation is prepared from:

Principles of VLSI RTL Design — Mike Gianfagna

CMOS VLSI Design, A Circuits and Systems Perspective — Neil Weste, David Harris





Agenda

Power Analysis

- # 1. Power Concept
- # 2. Power Saving Techniques
- # 3. Applying Power for MIPS Example

Agenda

Power Analysis

1. Power Concept
2. Power Saving Techniques
3. Applying Power for MIPS Example

1. Power Measurement on FPGA

Agenda

1. Power Measurement [Chapter 5: Power]

1.1 Power and Energy

1.2 Dynamic Power

1.3 Static Power

Agenda

- # 1. Power Measurement
  - # 1.1 Power and Energy
  - # 1.2 Dynamic Power
  - # 1.3 Static Power

Power and Energy

- Power is drawn from a voltage source attached to the V_{DD} pin(s) of a chip.
- Instantaneous Power:

$$P(t) =$$

- Energy:

$$E =$$

- Average Power:

$$P_{\text{avg}} =$$

Power in Circuit Elements

$$P_{VDD}(t) = I_{DD}(t) V_{DD}$$

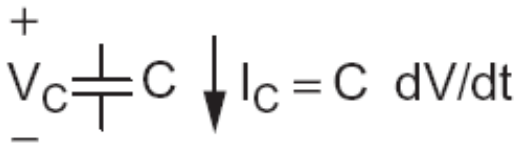


$$P_R(t) = \frac{V_R^2(t)}{R} = I_R^2(t) R$$



$$E_C = \int_0^{\infty} I(t) V(t)\, dt = \int_0^{\infty} C \frac{dV}{dt} V(t)\, dt$$

$$= C \int_0^{V_C} V(t)\, dV = \tfrac{1}{2} C V_C^2$$



Charging a Capacitor

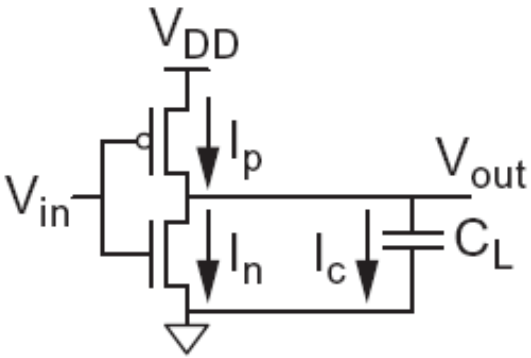
- When the gate output rises
  - Energy stored in capacitor is

$$E_C = \tfrac{1}{2} C_L V_{DD}^2$$

  - But energy drawn from the supply is

$$E_{VDD} = \int_0^\infty I(t) V_{DD} dt = \int_0^\infty C_L \frac{dV}{dt} V_{DD} dt$$

$$= C_L V_{DD} \int_0^{V_{DD}} dV = C_L V_{DD}^2$$



  - Half the energy from V_{DD} is dissipated in the pMOS transistor as heat, other half stored in capacitor
- When the gate output falls
  - Energy in capacitor is dumped to GND
  - Dissipated as heat in the nMOS transistor

Power Waveform

- Example: V_{DD} = 1.0 V, C_L = 150 fF, f = 1 GHz

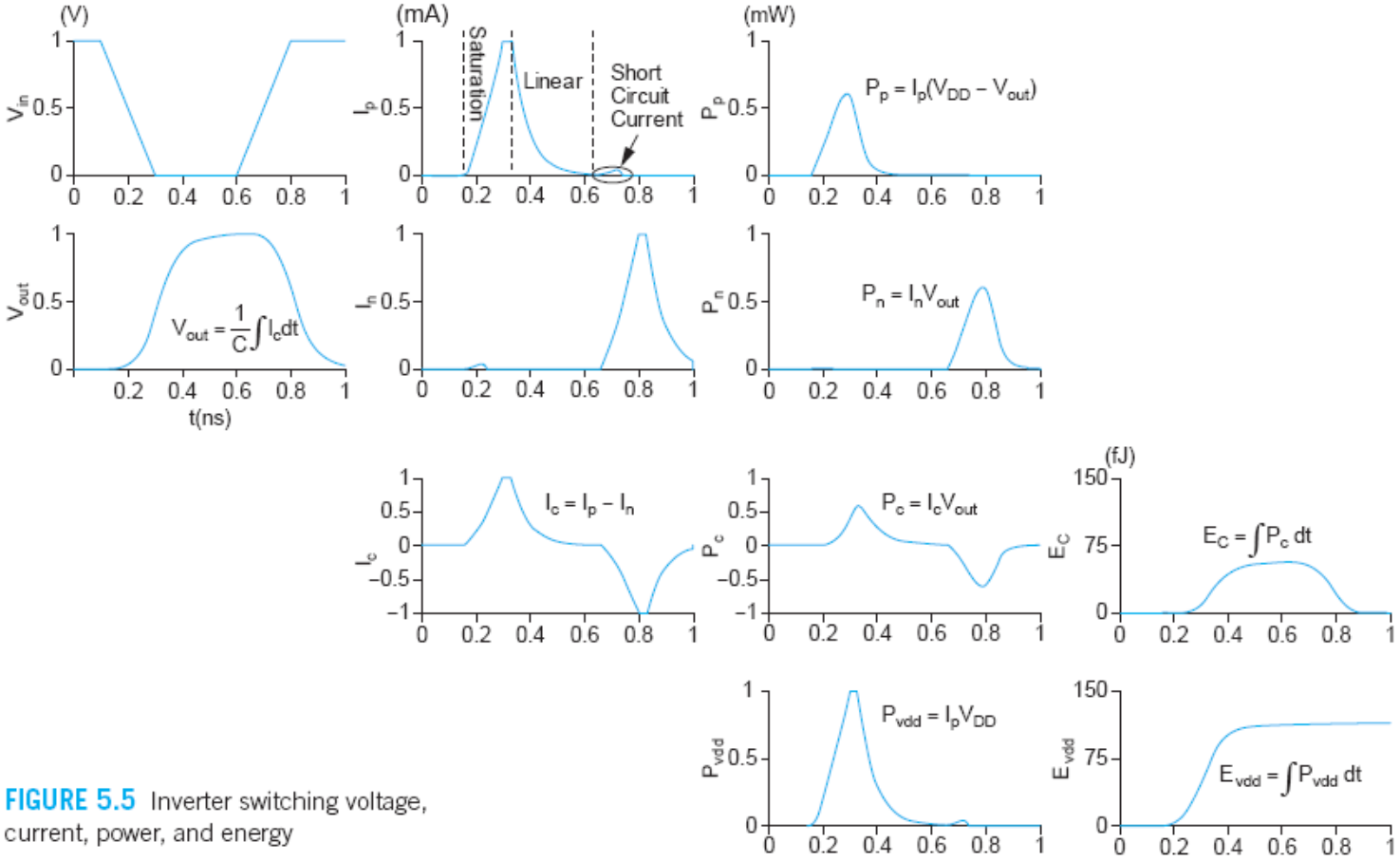
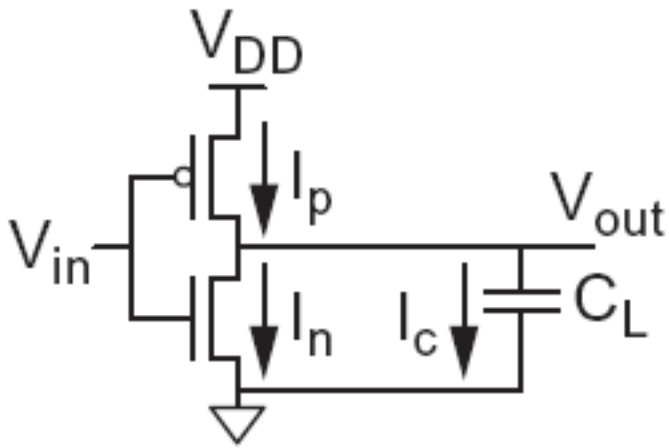


FIGURE 5.5 Inverter switching voltage, current, power, and energy



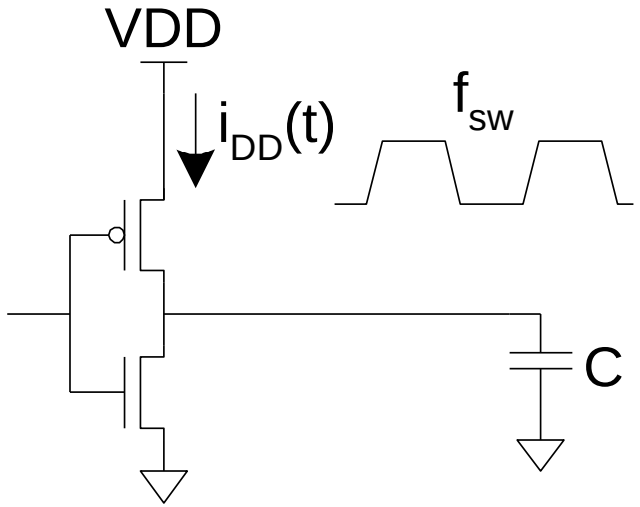
Switching Power

$$P_{\text{switching}} = \frac{1}{T}\int_0^T i_{DD}(t)V_{DD}\,dt$$

$$= \frac{V_{DD}}{T}\int_0^T i_{DD}(t)\,dt$$

$$= \frac{V_{DD}}{T}\left[Tf_{\text{sw}}CV_{DD}\right]$$

$$= CV_{DD}{}^2 f_{\text{sw}}$$



Activity Factor

- Suppose the system clock frequency = f
- Let $f_{sw} = \alpha f$, where α = activity factor
  - If the signal is a clock, $\alpha = 1$
  - If the signal switches once per cycle, $\alpha = ½$

- Dynamic power:

$$P_{\text{switching}} = \alpha C V_{DD}^{\ 2} f$$

Short Circuit Power

- When transistors switch, both nMOS and pMOS networks may be momentarily ON at once
- Leads to a blip of “short circuit” current.
- < 10% of dynamic power if rise/fall times are comparable for input and output
- We will generally ignore this component

Power Dissipation Sources

- $P_{total} = P_{dynamic} + P_{static}$

Dynamic power:

$$P_{dynamic} = P_{switching} + P_{shortcircuit}$$

- Switching load capacitances
- Short-circuit current

Static power:

$$P_{static} = (I_{sub} + I_{gate} + I_{junct} + I_{contention})V_{DD}$$

- Subthreshold leakage
- Gate leakage
- Junction leakage
- Contention current

Agenda

- # 1. Power Measurement
  - # 1.1 Power and Energy
  - # 1.2 Dynamic Power
  - # 1.3 Static Power

Dynamic Power Example

- 1 billion transistor chip
  - 50M logic transistors
    - Average width: 12 λ
    - Activity factor = 0.1
  - 950M memory transistors
    - Average width: 4 λ
    - Activity factor = 0.02
  - 1.0 V 65 nm process
  - C = 1 fF/μm (gate) + 0.8 fF/μm (diffusion)
- Estimate dynamic power consumption @ 1 GHz. Neglect wire capacitance and short-circuit current.

Solution

$$C_{\text{logic}} = \left(50 \times 10^6\right)\left(12\lambda\right)\left(0.025\mu m / \lambda\right)\left(1.8\, fF / \mu m\right) = 27 \text{ nF}$$

$$C_{\text{mem}} = \left(950 \times 10^6\right)\left(4\lambda\right)\left(0.025\mu m / \lambda\right)\left(1.8\, fF / \mu m\right) = 171 \text{ nF}$$

$$P_{\text{dynamic}} = \left[0.1C_{\text{logic}} + 0.02C_{\text{mem}}\right](1.0)^2\,(1.0 \text{ GHz}) = 6.1 \text{ W}$$

Dynamic Power Reduction

$$P_{\text{switching}} = \alpha C V_{DD}{}^2 f$$

- Try to minimize:
  - Activity factor
  - Capacitance
  - Supply voltage
  - Frequency

1. Activity Factor Reduction

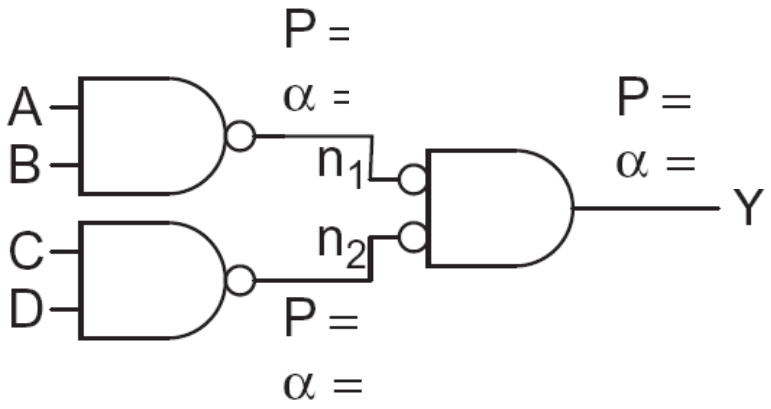
- Let P_i = Prob(node i = 1)
  - $\overline{P}_i = 1-P_i$
- $\alpha_i = P_i * \overline{P}_i$
- Completely random data has P = 0.5 and $\alpha = 0.25$
- Data is often not completely random
  - e.g. upper bits of 64-bit words representing bank account balances are usually 0
- Data propagating through ANDs and ORs has lower activity factor
  - Depends on design, but typically $\alpha \approx 0.1$

Switching Probability

Gate	P_Y
AND2	$P_A P_B$
AND3	$P_A P_B P_C$
OR2	$1 - \overline{P}_A \overline{P}_B$
NAND2	$1 - P_A P_B$
NOR2	$\overline{P}_A \overline{P}_B$
XOR2	$P_A \overline{P}_B + \overline{P}_A P_B$

Example

- A 4-input AND is built out of two levels of gates
- Estimate the activity factor at each node if the inputs have P = 0.5



Agenda

- # 1. Power Measurement
  - # 1.1 Power and Energy
  - # 1.2 Dynamic Power
  - # 1.3 Static Power

Static Power

- Static power is consumed even when chip is quiescent.
  - Leakage draws power from nominally OFF devices
  - Ratioed circuits burn power in fight between ON transistors

Static Power Example

- Revisit power estimation for 1 billion transistor chip
- Estimate static power consumption
  - Subthreshold leakage
    - Normal V_t: 100 nA/μm
    - High V_t: 10 nA/μm
    - High Vt used in all memories and in 95% of logic gates
  - Gate leakage 5 nA/μm
  - Junction leakage negligible

Solution

$$W_{\text{normal-V}_t} = \left(50\times10^6\right)\left(12\lambda\right)\left(0.025\mu\text{m}/\lambda\right)\left(0.05\right) = 0.75\times10^6\ \mu\text{m}$$

$$W_{\text{high-V}_t} = \left[\left(50\times10^6\right)\left(12\lambda\right)\left(0.95\right)+\left(950\times10^6\right)\left(4\lambda\right)\right]\left(0.025\mu\text{m}/\lambda\right) = 109.25\times10^6\ \mu\text{m}$$

$$I_{sub} = \left[W_{\text{normal-V}_t}\times100\ \text{nA}/\mu\text{m}+W_{\text{high-V}_t}\times10\ \text{nA}/\mu\text{m}\right]/2 = 584\ \text{mA}$$

$$I_{gate} = \left[\left(W_{\text{normal-V}_t}+W_{\text{high-V}_t}\right)\times5\ \text{nA}/\mu\text{m}\right]/2 = 275\ \text{mA}$$

$$\text{P}_{static} = \left(584\ \text{mA}+275\ \text{mA}\right)\left(1.0\ \text{V}\right) = 859\ \text{mW}$$

Agenda

Power Analysis

1. Power Concept

2. Power Saving Techniques

3. Applying Power for MIPS Example

Agenda

2. Power Saving Techniques

- # 2.1 Switching Activity
- # 2.2 Supply Voltage
- # 2.3 Selective Shut Down
- # 2.4 Load Capacitance
- # 2.5 Input Transition
- # 2.6 Device Characteristics

2.1 Switching Activity

- Switching Activity represents the times there is a transition at an input.
- As it increases, the times when the instantaneous short circuit is established will be greater. Then, the short-circuit power increases.
- Also, as it increases, the output switches increase, then the number of charging and discharging of the load capacitance will be higher.
- Some common techniques used for reducing Switching Activity will be explained.

2.1 Switching Activity

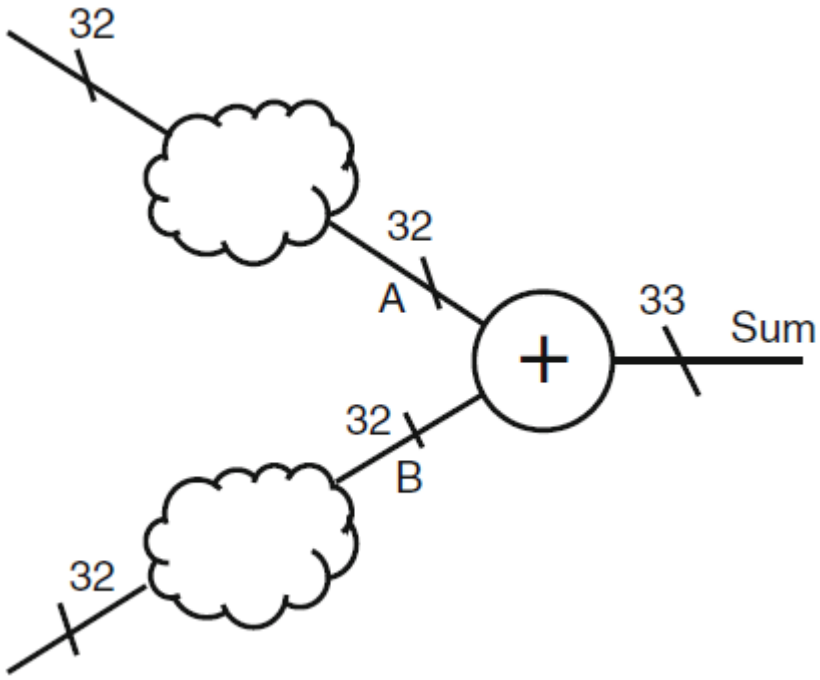
1. **Shifter Instead of Multipliers**

- In an XOR gate, any transition at any of the inputs will always cause a transition at the output. So, XOR gate is more power hungry.
- Multipliers (and adders) are made of a large number of XORs. So, even a single transition at any input will result in a lot of transitions internal to the multipliers.
- There are certain multiplication values, where the same result can be achieved using another operation.
- The multiplier by 2, 4, 8 etc. should be replaced by a Left Shifter. Similarly, some division by 2, 4, 8 etc. can be replaced by a Right Shifter.

2.1 Switching Activity

2. Operand Isolation

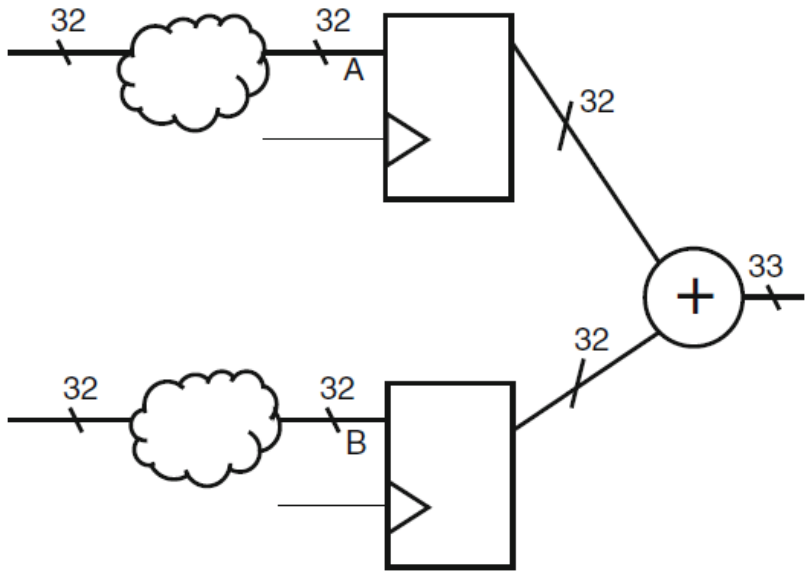
- All the 32 bits of the busses A or B will not arrive at the adder at the same instant, as they arrive through different combinational gates or different routing tracks.
- As each signal at the input of the adder changes, the adder reevaluates and the output switches.
- All the intermediate results were of no use, but they consumed power.
- Arithmetic operators such as Adders and Multipliers have lots of XORs, causing more switching activity down the circuit.



2.1 Switching Activity

2. Operand Isolation

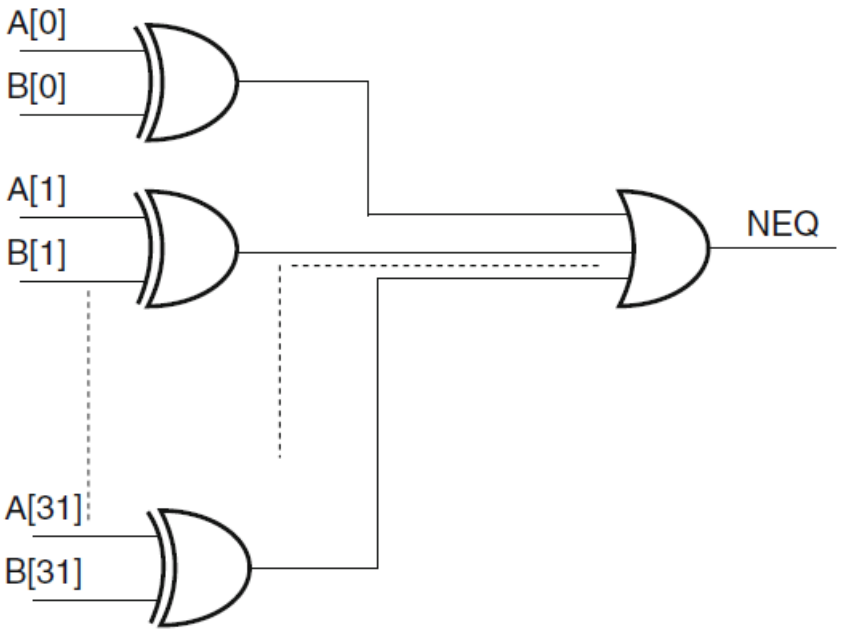
- This can be done by registering all the inputs, then, feeding the output directly to the adder, avoiding the intermediate switching.
- Since all the operands are now coming directly from the registers, they will reach the operator at almost the same time.
- The cost of using this technique is adding some area (for the additional registers). Also, there is an increase (by one clock cycle) in latency.



2.1 Switching Activity

3. **Avoid Comparison**

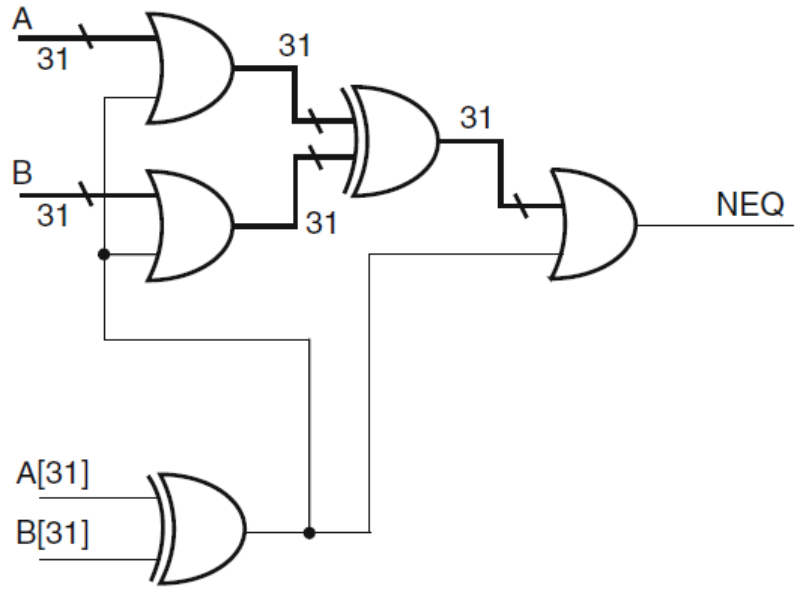
- Comparator operations also involve XORs.
- If there is a change in any of the inputs, there will be lot of intermediate transitions (glitch) while the final output may not be changed.
- These intermediate transitions also consume power (glitch power).
- In the context of a comparator, statistically, comparing just the MSB would be able to give the results in 50% of the cases.



2.1 Switching Activity

3. Avoid Comparison

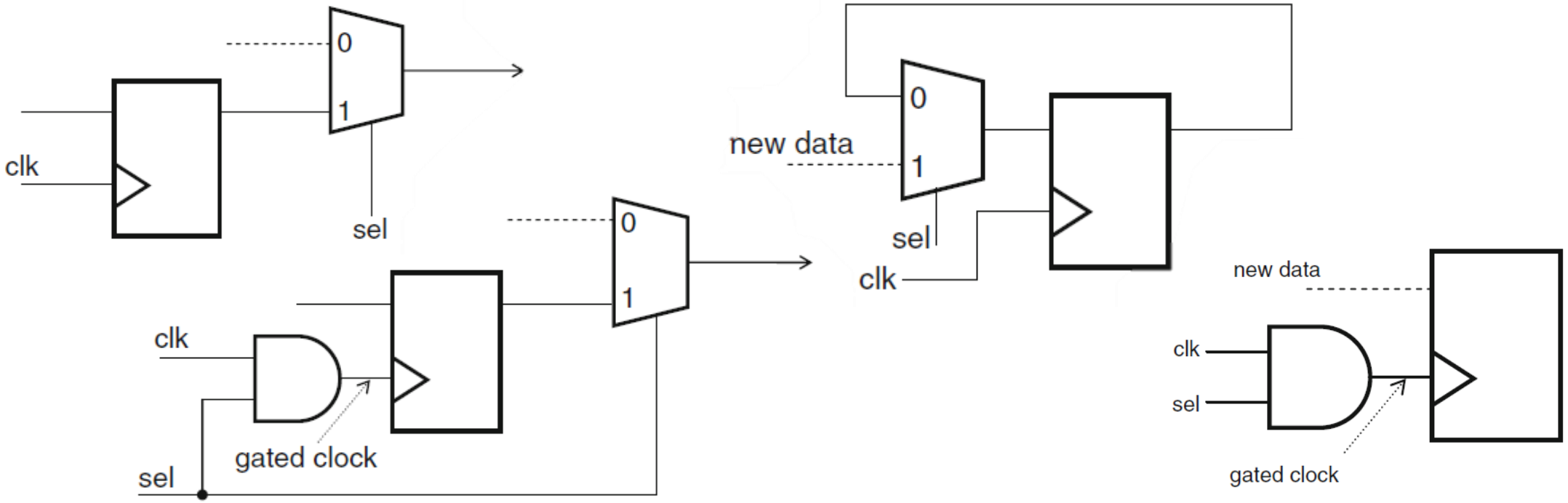
- The lower 31 bits should reach the XOR only after the MSBs have been evaluated, otherwise they would create the glitch again.
- The same concept can be further applied recursively – for the comparison of the lower 31 bits also. And so on.
- The cost of doing this modification is:
  - The additional required area.
  - The paths timing get modified.
  - Every time the change in the MSB will cause additional switching power.



2.1 Switching Activity

4. Clock Gating

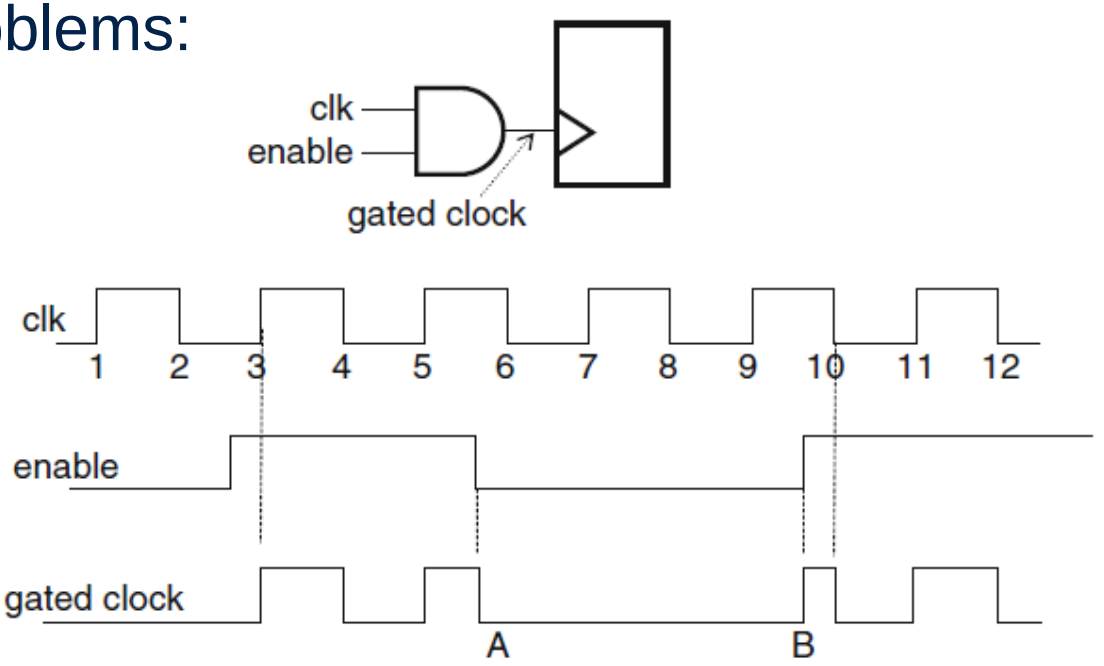
- Clock gating is the most popular and most effective method to reduce switching rate.
- The data need not be captured on the flop when output is not being sampled (not used), or the input is not changing.



2.1 Switching Activity

4. Clock Gating - Pulse Clipping, Spurious Clocking

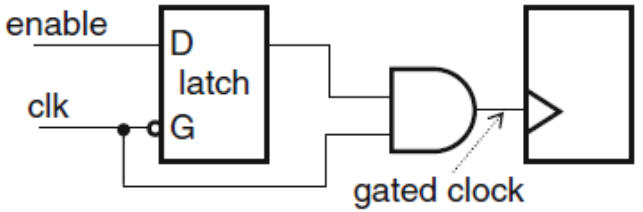
- The falling edge which should have come at 6 has come earlier.
- Effectively, the High pulse of the clock has been shortened (pulse clipping), resulting in Pulse width Violation.
- There is a not-needed positive edge at point B on the gated clock (spurious clocking), resulting in two possible problems:
- A glitch could get captured in the flop, causing a Hold-Time issue.
- The next stage does not get one complete cycle for the (glitch) value, causing a setup-time issue at the next stage.

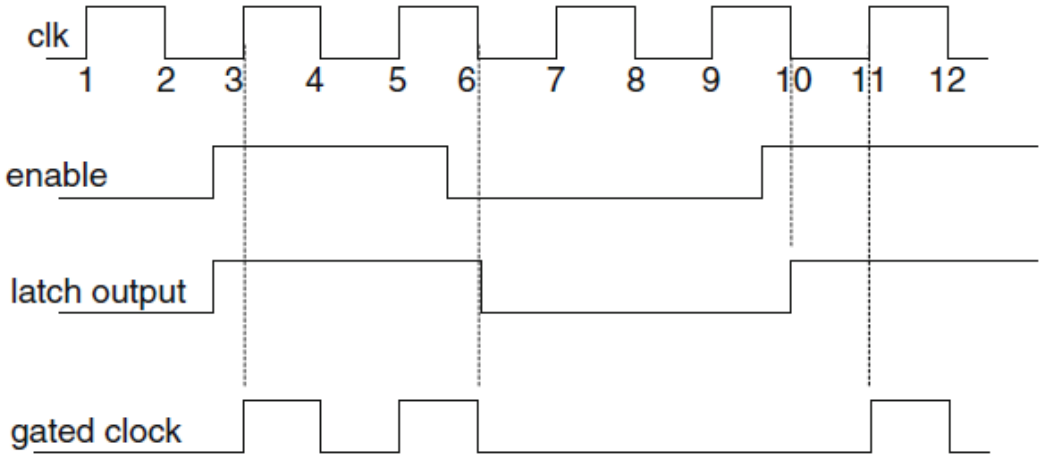


2.1 Switching Activity

4. Clock Gating - Pulse Clipping, Spurious Clocking

- To avoid the previous issues, the enable signal should be received on the gate only when the clock is Low.
- The Low latch will block enable's transitions from reaching the AND gate, till the clock has become Low, ensuring that the AND gate will see a transition on enable, only when the clock is low.

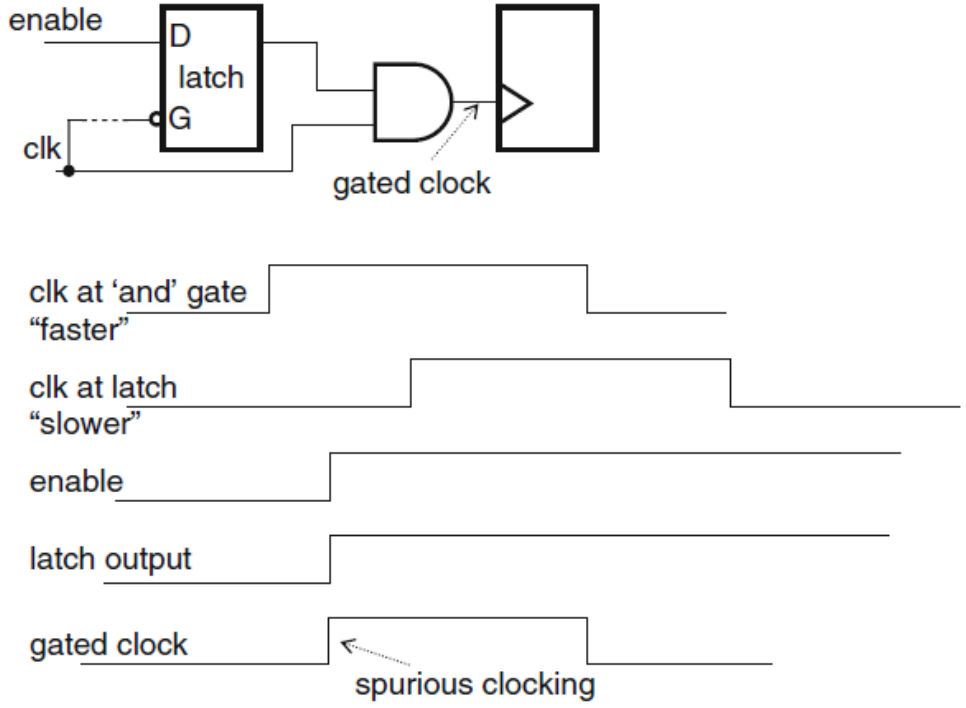


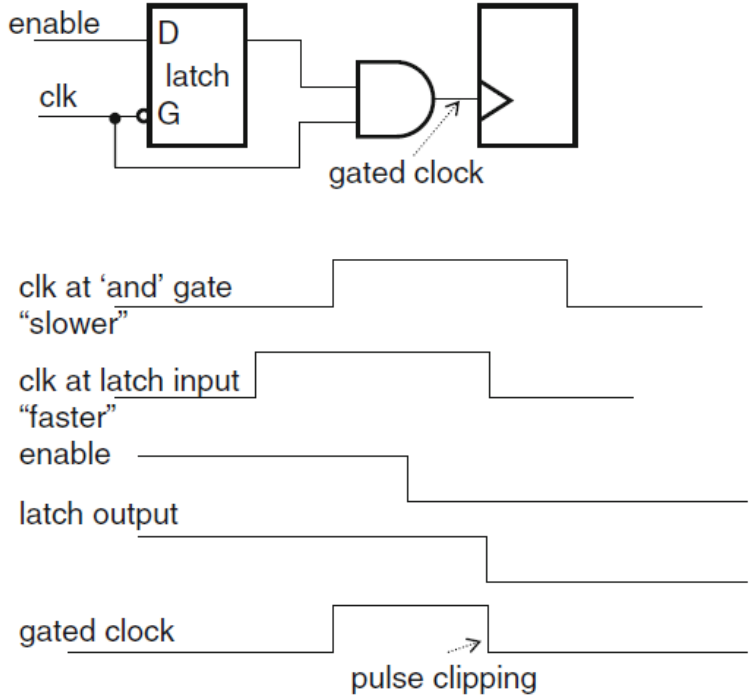


2.1 Switching Activity

4. Clock Gating - ICG

- Due to differential routing delays, it so happens that the clock to the AND gate path is different than the clock-to-latch path.
- To prevent these delays, the latch and the AND gate are put together in a cell with balanced path-delays (Integrated Clock Gating cell, ICG cell).

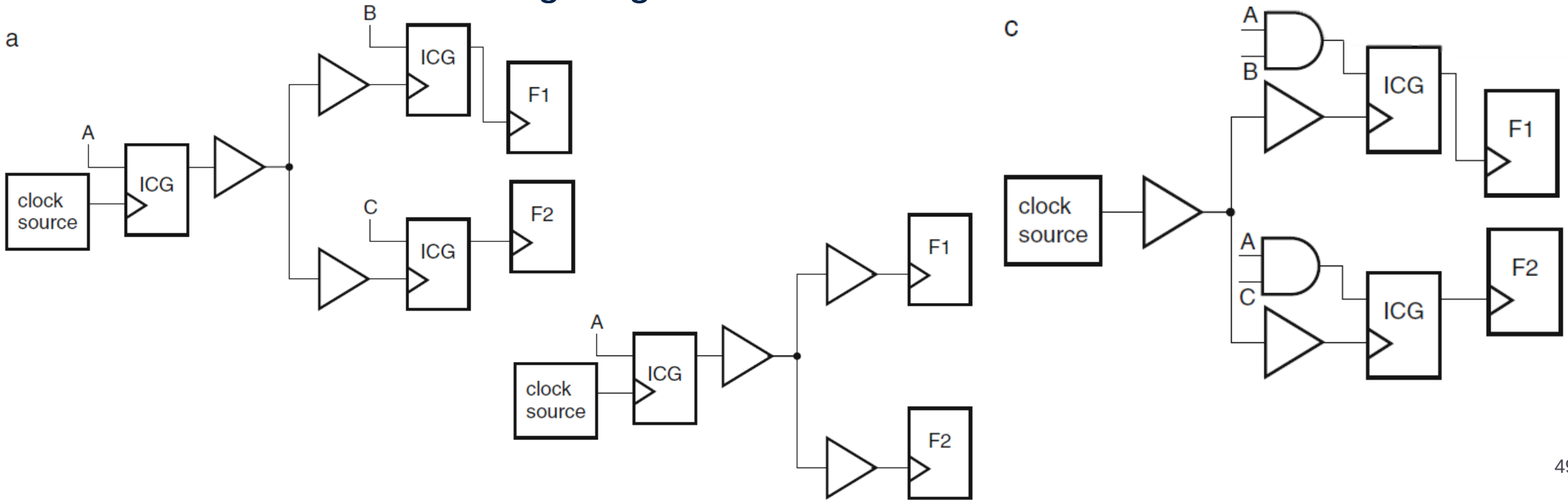




2.1 Switching Activity

4. Clock Gating - Location

- The closer to the source you do the gating, the more power you can save.
- The choice depends on the number of impacted flops and the overall duration – for which the gating circuits block the clocks.

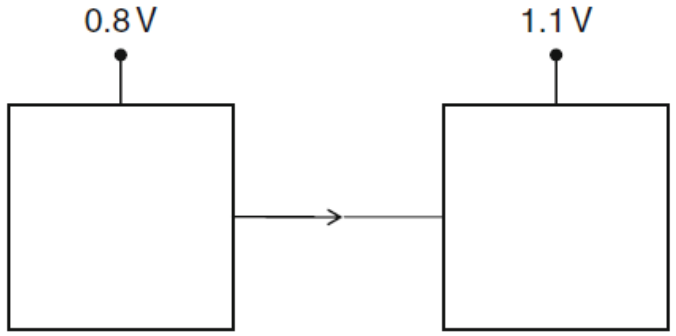


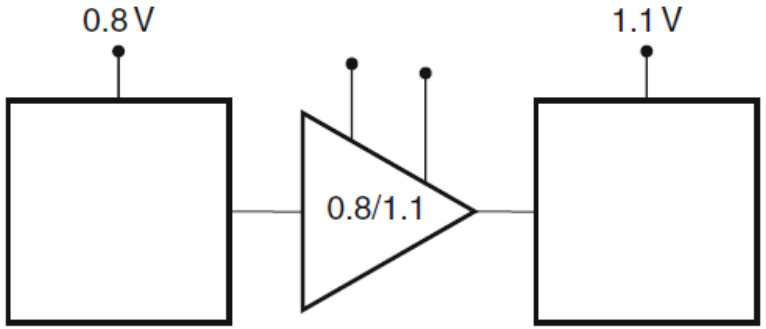
2.2 Supply Voltage

- The charge stored on the capacitive load is proportional to the square of supply voltage. So, higher supply voltage means. greater charge stored on the capacitor during charging. So, switching power is proportional to the square of supply voltage.
- The transistors in the short-circuit path behave like resistive components. Then, the flowing current depends on the supply voltage, proportional to the square of the Supply Voltage.
- The leakage current flowing through the reverse biased junctions (i.e. OFF transistors) is dependent on the Supply Voltage.

2.2 Supply Voltage

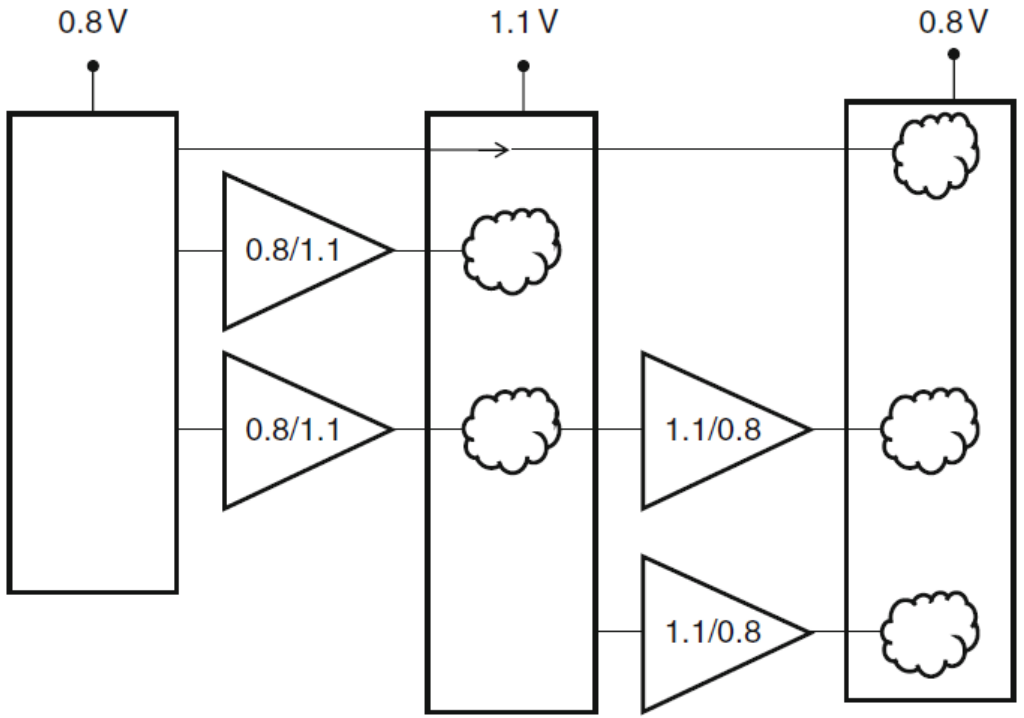
- The technology should support the devices operating at a lower voltage.
- Multiple voltage domains might be used. Components on the critical path need to work at higher voltage, while the other components with enough timing slack use lower voltage.
- When a High Signal at 0.8 V signal reaches 1.1 V domain, the noise-margin is reduced to just 0.15 V (0.8 – 0.65)
- A level shifter is put when a signal moves among voltage domains.





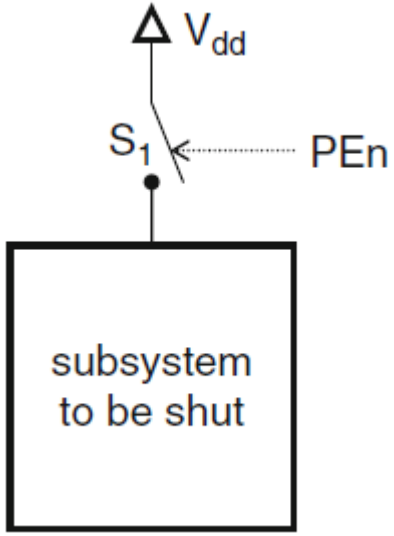
2.2 Supply Voltage

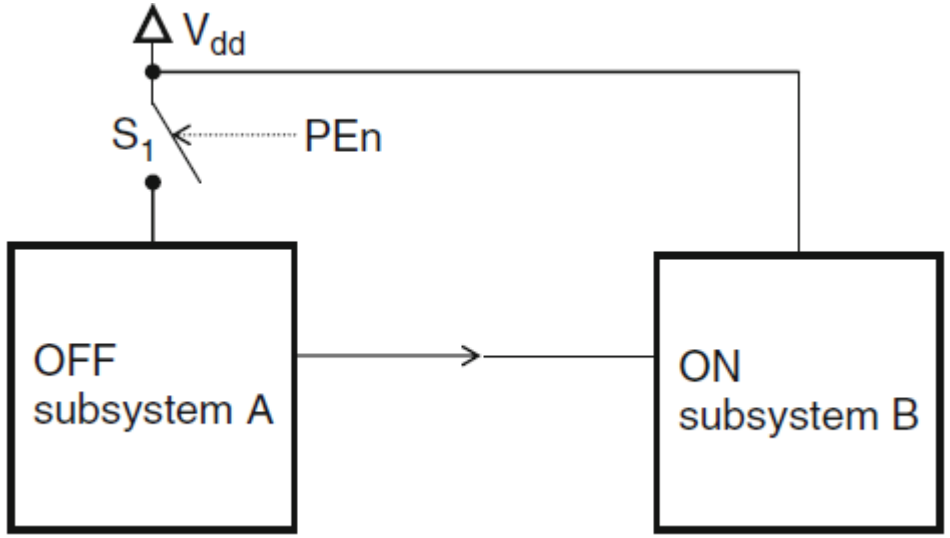
- For each combination of source and destination voltage levels a different kind of shifter is used.



2.3 Selective Shut Down

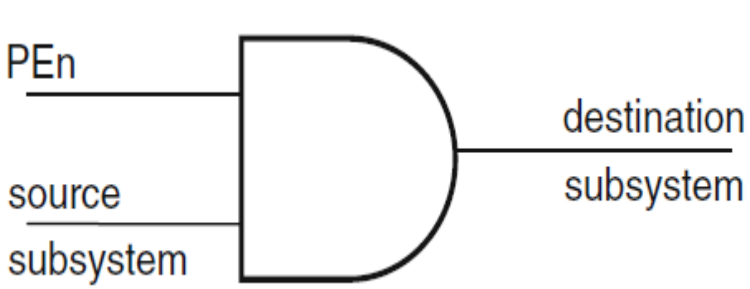
- Portions of the circuit which are not in use can be selectively shut down.
- Leakage power is the only savings through selective shut down since there is no activity anyways.
- The power supply to a portion of the circuit which is expected to be non-active for long durations might be switched off, during such durations.

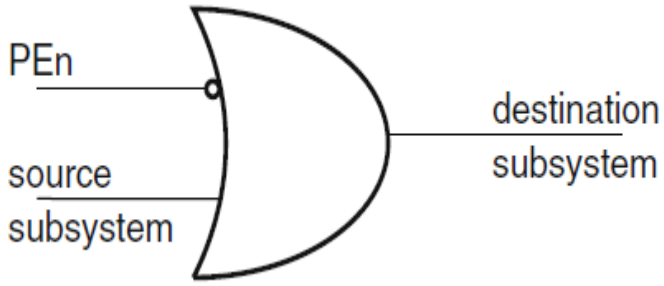


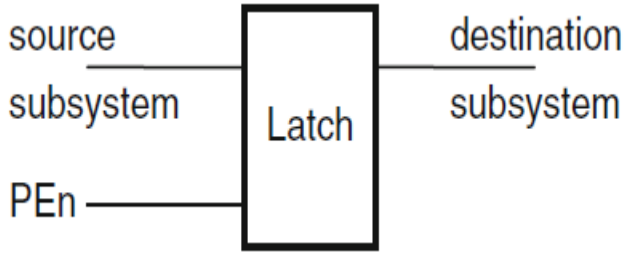


2.3 Selective Shut Down

- Isolation between two subsystems is important as the input from A (assume OFF now) to B may be floating (may neither be at 0 or at 1, it may be in-between), causing a direct current path from supply to the ground(the device may be burned).
- There are 3 kinds of Isolation Cells.
  1. AND Isolation Cell: sends the src when Powered and 0 during Power Down.
  2. OR Isolation Cell: sends the src when Powered and 1 during Power Down.
  3. Latch Isolation Cell: sends the src when Powered and keep the preceding value during Power Down.

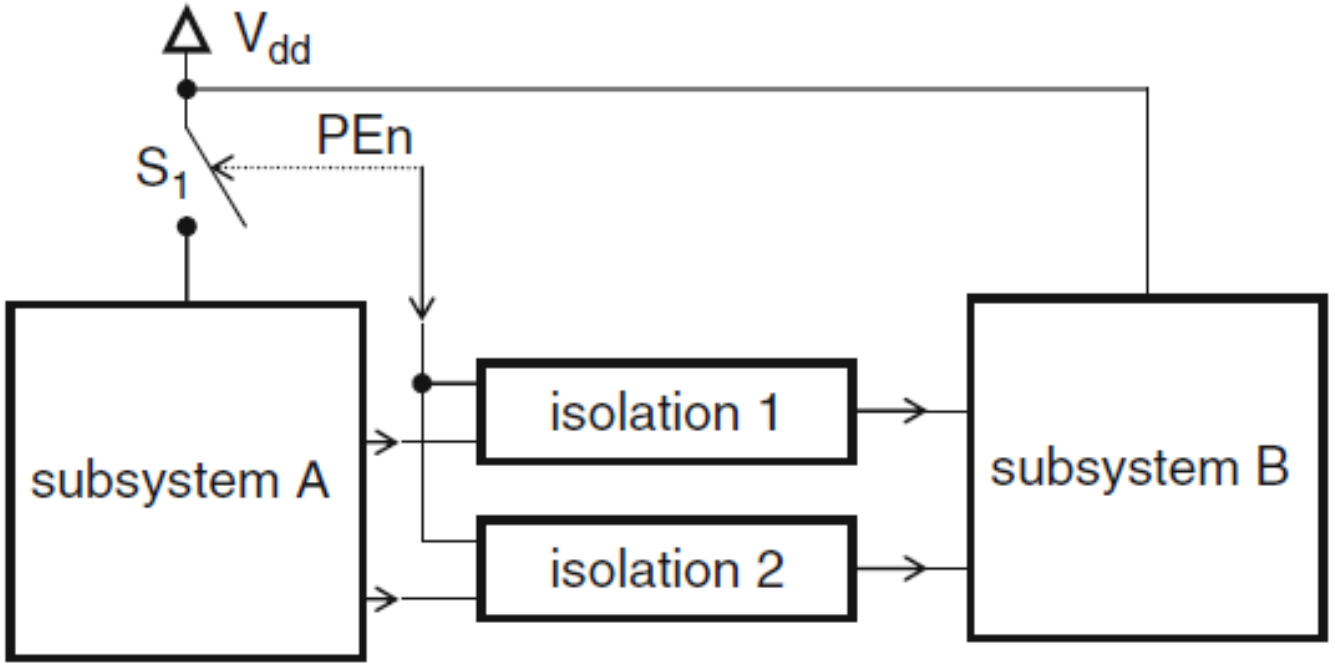






2.3 Selective Shut Down

- A complete hookup of the interaction among the 2 sub-systems.
- It is possible to use different types of Isolation Cells for the different signals.
- It is very important that the logic generating the PowerEnable signal should be Always-On.



Agenda

Power Analysis

1. Power Concept

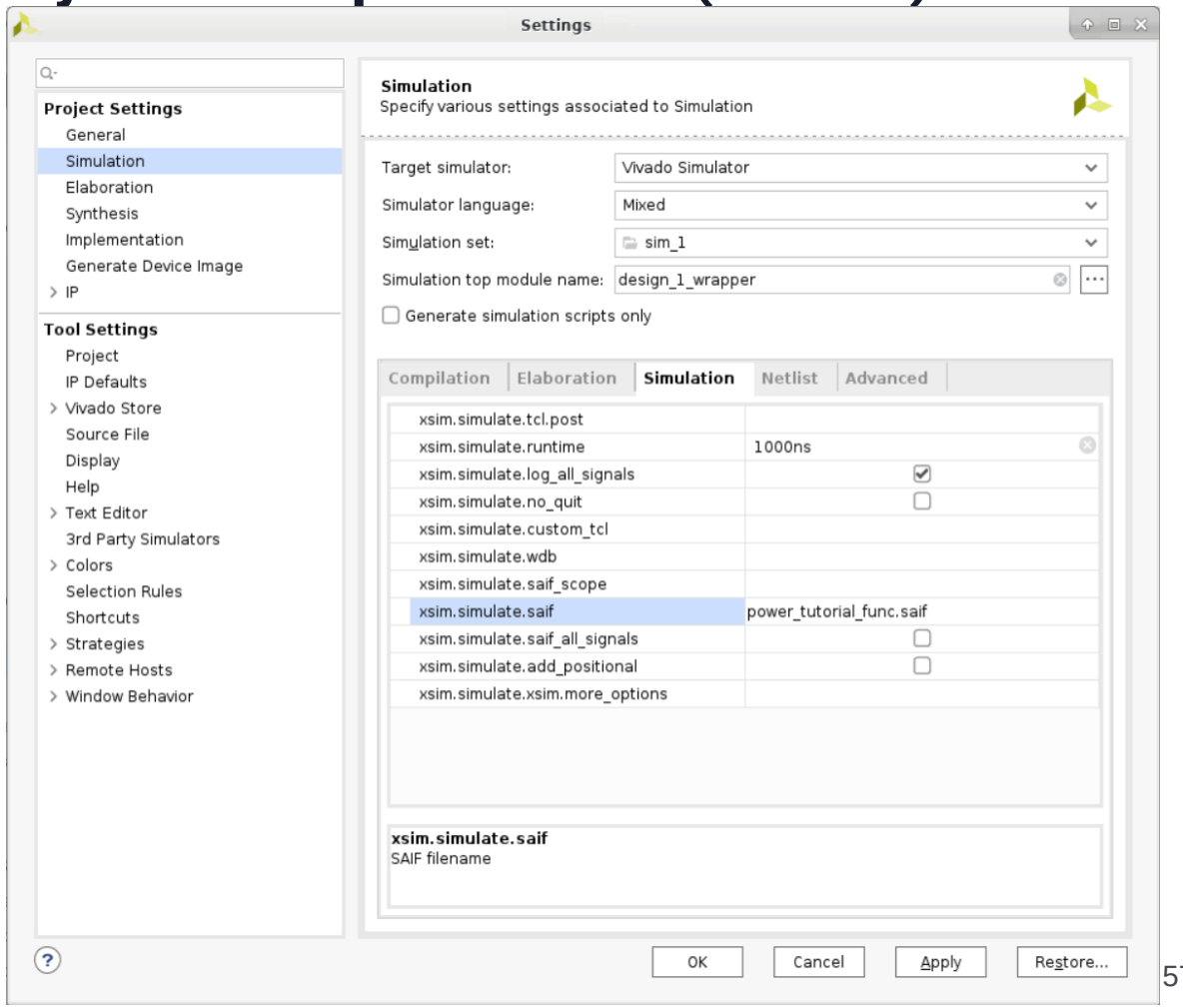
2. Power Saving Techniques

3. Applying Power for MIPS Example

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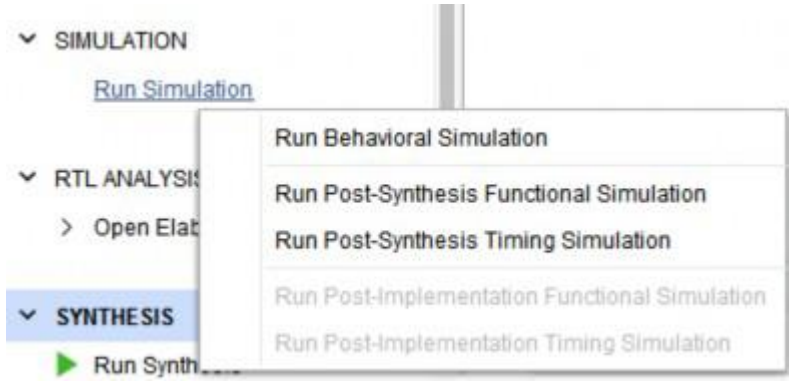
Vivado Design Suite Tutorial: Power Analysis and Optimization (UG997)

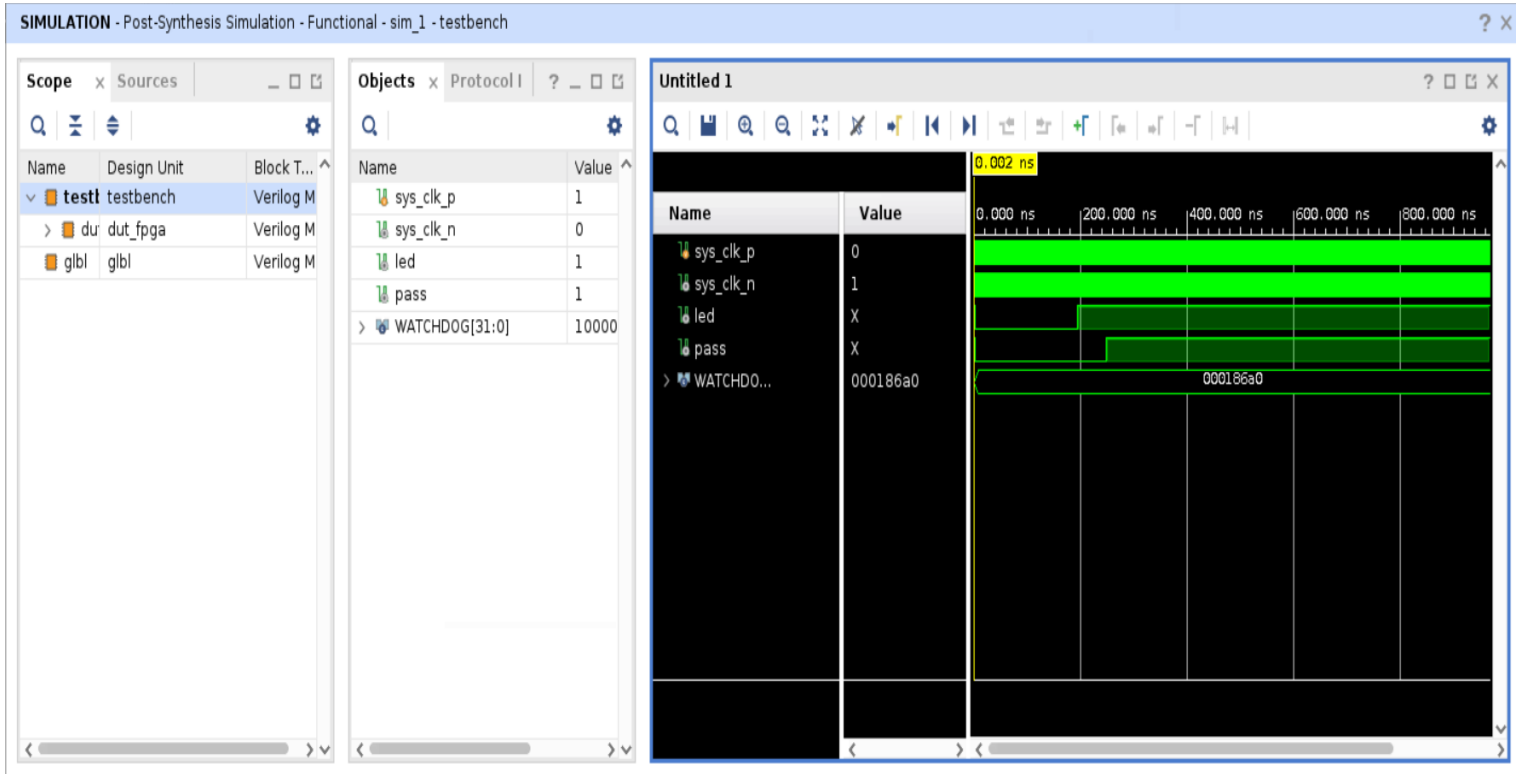
- From the navigator at the left, press “Settings”.
- A widow will open, select “Simulation” from the navigator at the left.
- Then, In the Simulation tab, enter the SAIF file name as mips_power.saif for xsim.simulate.saif.
- Make sure to check the xsim.simulate.log_all_signals box.
- Click OK.



3. Applying Power for MIPS Example

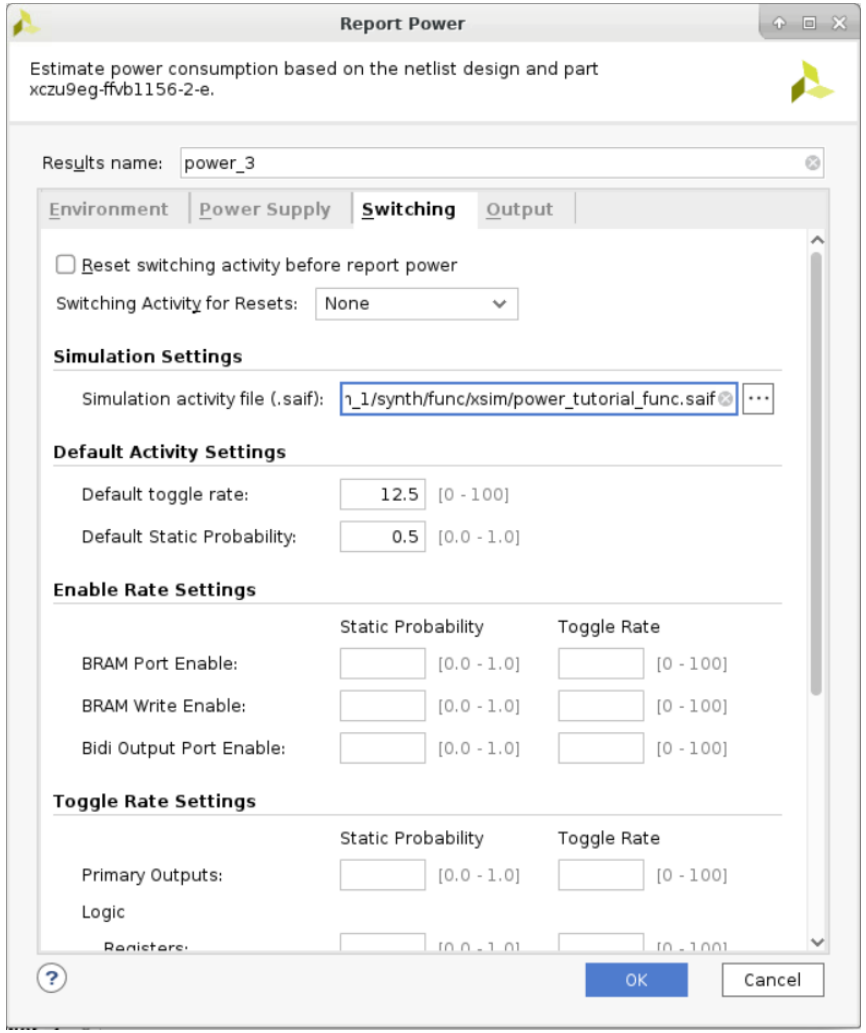
In the Flow Navigator, click Run Simulation > Run Post-Synthesis Functional Simulation.
When you launch the Run Post-Synthesis Functional Simulation command, the Vivado simulator is invoked to run the simulation





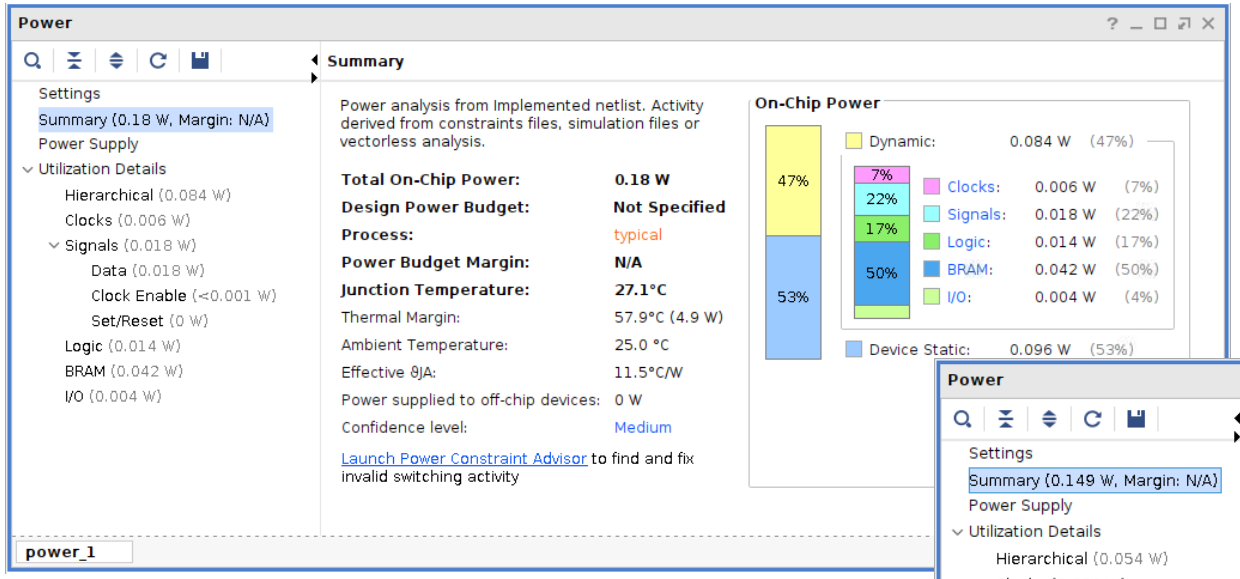
3. Applying Power for MIPS Example

- Ensure that the requested SAIF file is generated.
- In the Flow Navigator window, click Open Synthesized Design to expand options.
- From the Synthesized Design options, select **Report Power**.
- In the **Output** tab of Report Power dialog box, make the following changes:
  - Set the Output text File to power_3.pwr
  - Set the Output XPE File to power_3.xpe
- In the **Environment** tab of Report Power dialog box, make sure that the Process is set to maximum.
- In the **Switching** tab of Report Power dialog box, specify the SAIF file location.



3. Applying Power for MIPS Example

- With default Switching Activity



- With Switching Activity of Test1

